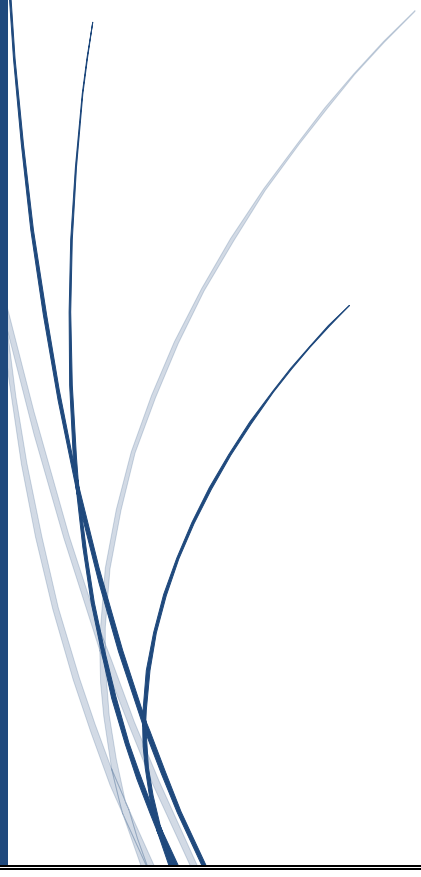
ANALOG DEVICES

AHEAD OF WHAT'S POSSIBLE™

AHB Master Interface

ADI Digital Design Internship

By: **Youssef Khaled**



1. Supported Features

1) Pipelined Operation

- Implements address phase and data phase pipelining, enabling the initiation of a new transfer while the previous transfer completes.

2) Burst Transaction Support

- Supports SINGLE and INCR (undefined length) burst transactions.
- For INCR bursts, the address is automatically incremented according to the size defined by **HSIZE** during each sequential phase.

3) Configurable Data Widths

- Supports Byte (8-bit), Halfword (16-bit), and Word (32-bit) transfers.

4) Wait State Handling

- The master checks the **HREADY** signal before updating internal control lines or advancing state.
- Ensures data and address are only driven onto the bus when the slave is ready, avoiding protocol violations.

5) Busy Master Capability

- Supports a busy state insertion by setting whenever **BUSY_IN** is asserted.
- Allows delaying a transaction without returning to IDLE, as per AHB protocol.

6) External Control Interface

- Address, data and transfer initiation and termination are provided by external logic representing the master side.

2. Unsupported features

1) **Error response handling**
2) **Protection control**
3) **Locked transfer**
4) **increment and wrapping burst types**
5) **Multi-layer AHB lite**
6) **Large bus widths size (double word, 4-word line, 8-word line and 1024 bit transfer)**

3. Hardware Architecture

In addition to the AHB signals described in the standard we have added additional ports that act as the master side, these ports are described below.

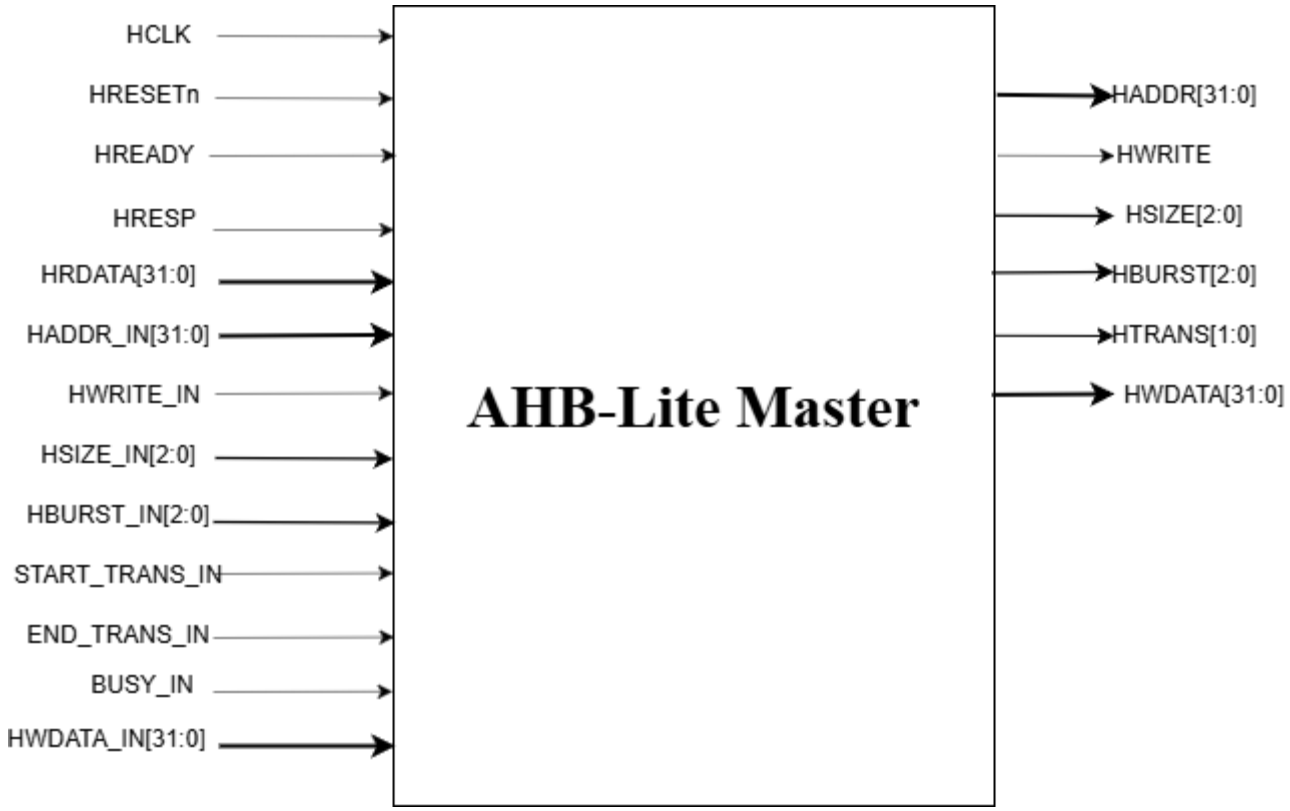


Figure 1: Master interface

- **HADDR_IN:** it represents the address coming from the master side.
- **HWRITE_IN:** it represents the operation the master wants to do (0 for read and 1 for write).
- **HBURST_IN:** it represents the burst type (SINGLE or INCR).
- **START_TRANS_IN:** it indicates that the master wants to start a new transaction, it must remain asserted during the whole transaction.
- **END_TRANS_IN:** it indicates that the master wants to terminate the transaction, it was added here especially to end the undefined length burst transactions.
- **BUSY_IN:** when HIGH, it indicates that the master is busy and hence the new **HTRANS** will be BUSY.
- **HWDATA_IN:** write data coming from master side.

4. FSM

The master FSM and its transitions and their description is given below.

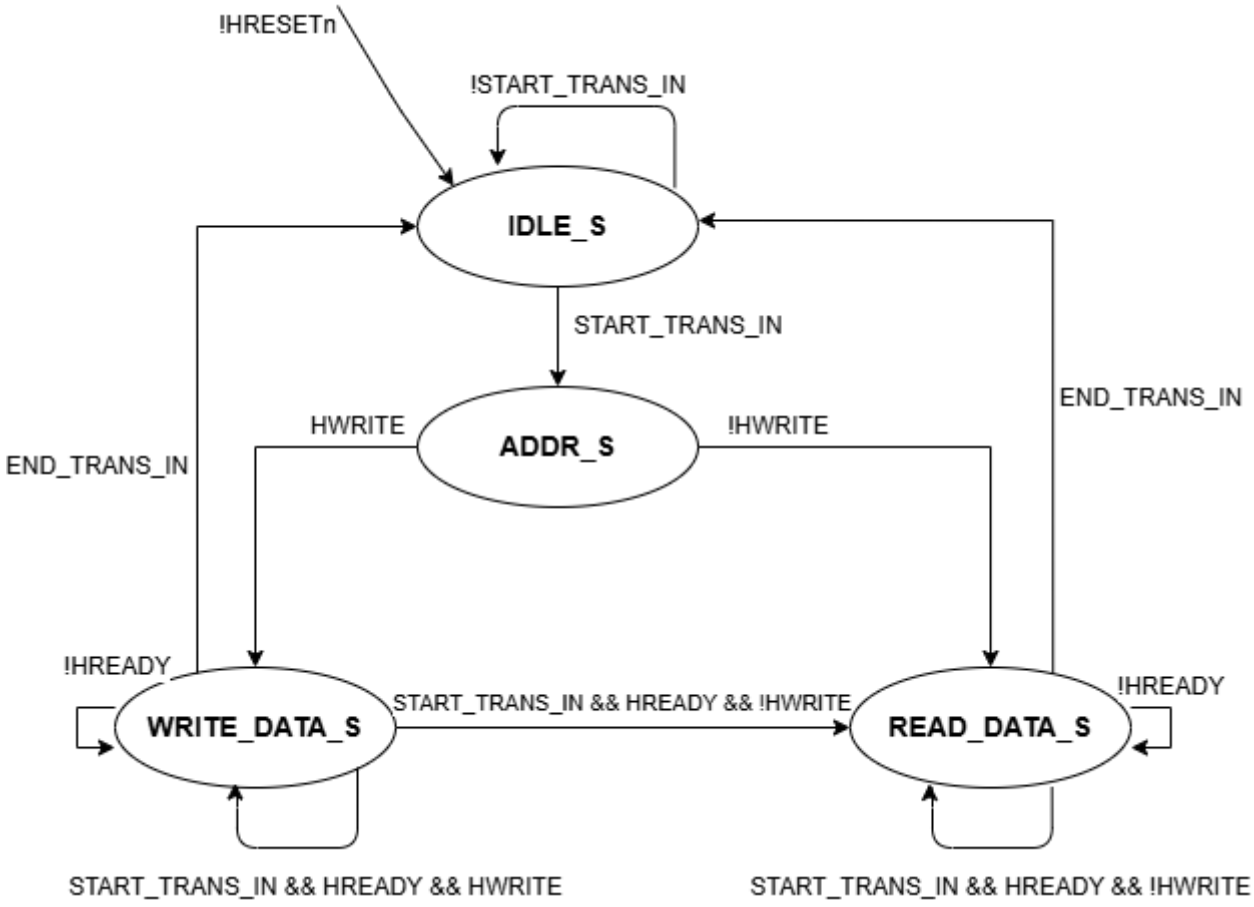


Figure 2: Master FSM

1) IDLE_S

- This is the IDLE state, which is the default state upon reset, during this state and according to the standard the address and control information must have valid levels, so here we initialized them.
- When START_TRANS_IN is asserted this indicates that master wants to initiate a new transaction so we go from IDLE_S to ADDR_S, else we stay in IDLE_S.
- HTRANS here is IDLE.

2) ADDR_S

- It represents the address phase, in which master puts the address and control information (transfer size, read or write operation, burst type).
- After a one clock cycle we go to another state depending on HWRITE signal's value, if it's LOW we go to DATA_READ_S and if it's HIGH we go to DATA_WRITE_S.
- HTRANS here is NONSEQ.

3) DATA_WRITE_S

- Here the master puts the write data (HWDATA) on the bus, and it must keep it stable (along with the address and control information) if the slave is not ready.
- Depending on burst type HADDR is handle, so for SINGLE burst we simply make the HADDR equals to HADDR_IN which is the address coming from the master side, but for INCR burst we make it equals to HADDR + 2^{HSIZE}.
- HTRANS here depends on either the burst ends or not so if the burst is INCR type we keep HTRANS seq till END_TRANS_IN asserted indicating end of the burst,

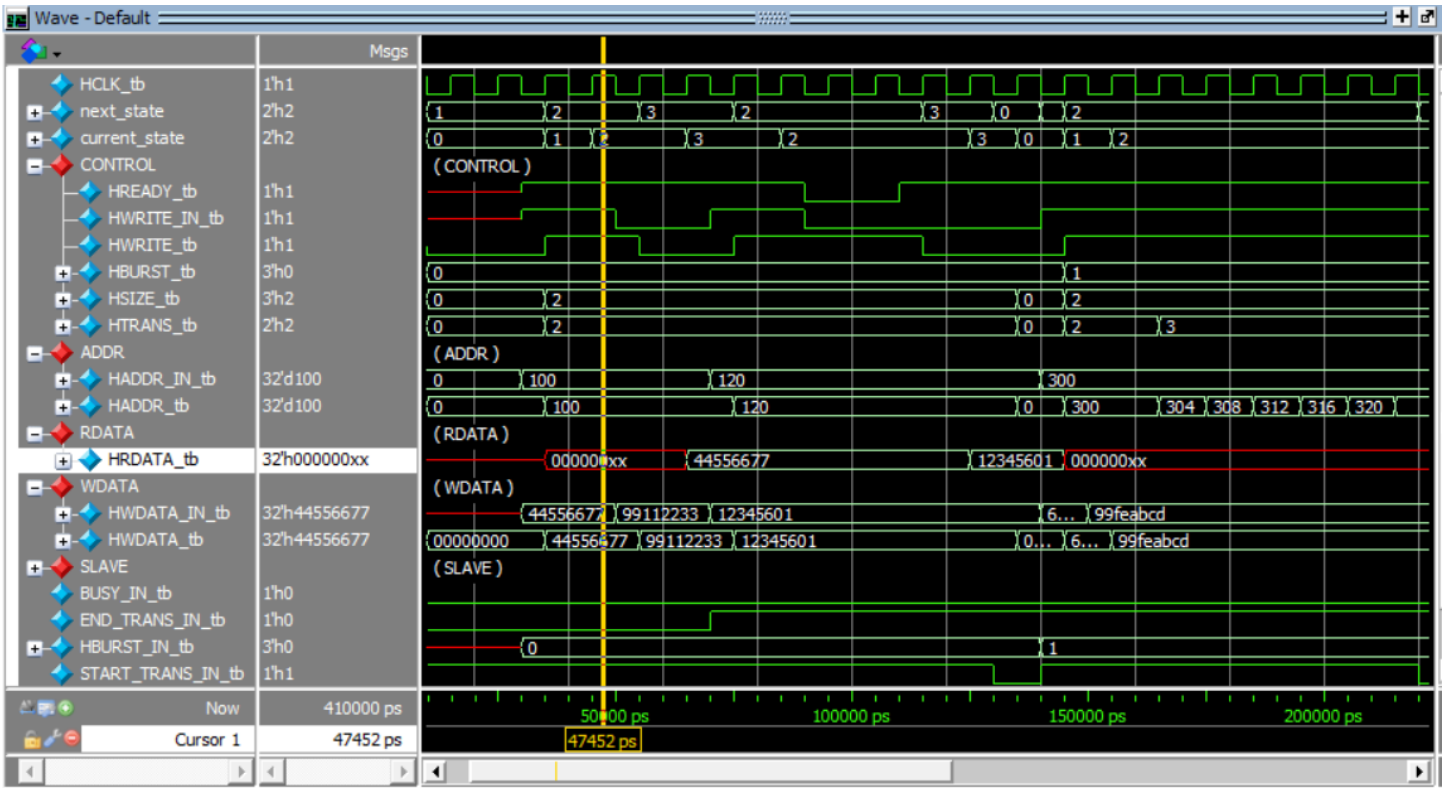
4) DATA_READ_S

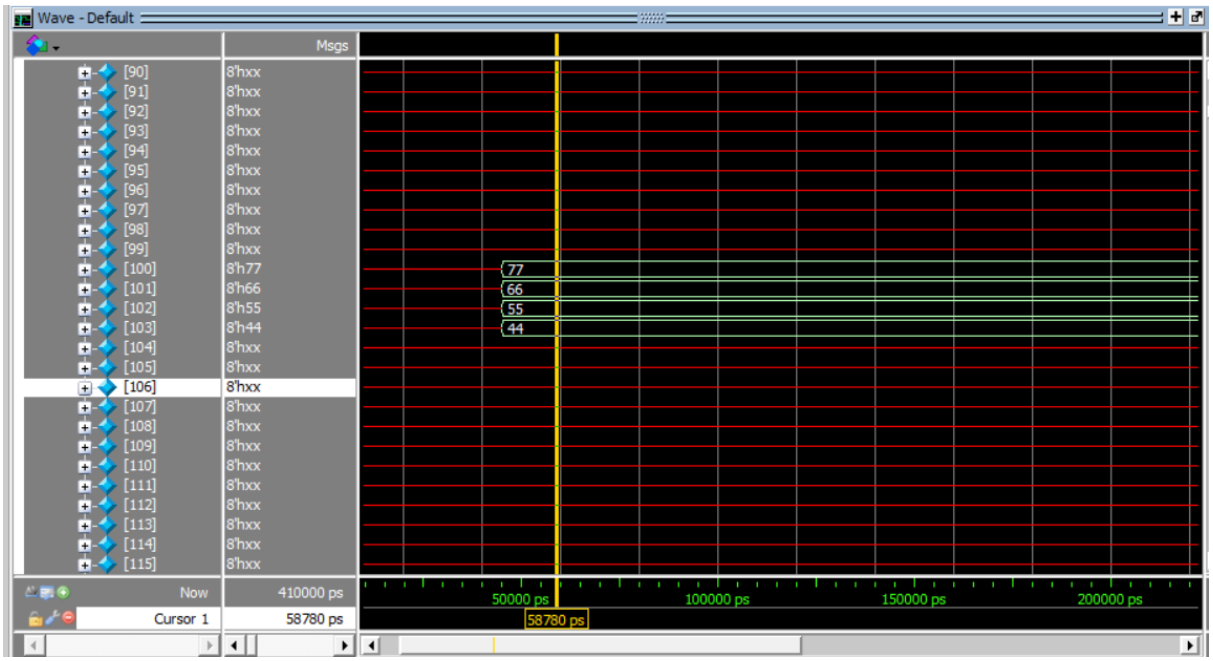
- Here the master waits for the slave to generate the read data HRDATA, it waits until HREADY is high, if HREAD is still low the address and control information must remain stable.
- Depending on burst type HADDR is handle, so for SINGLE burst we simply make the HADDR equals to HADDR_IN which is the address coming from the master side, but for INCR burst we make it equals to HADDR + 2^{HSIZE} (same as DATA_WRITE_S)
- HTRANS here depends on either the burst ends or not so if the burst is INCR type we keep HTRANS seq till END_TRANS_IN asserted indicating end of the burst. (same as DATA_WRITE_S)

5. Testing

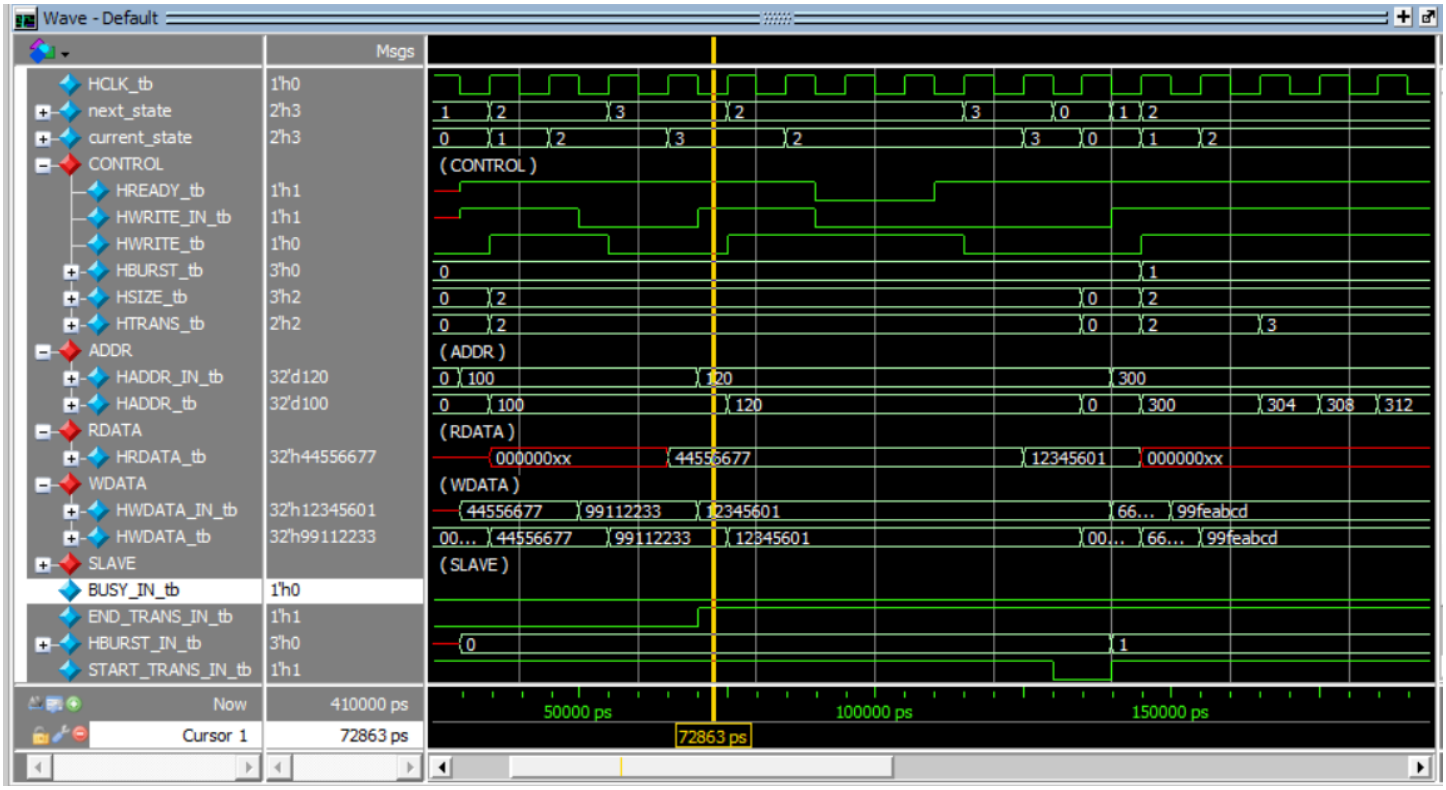
A simple directed testbench is used to test the master, where we test some cases described below. We emulate the behavior of the slave in the testbench by definition of a simple 2KB memory that reads from HRDATA and writes with HWDATA when ready.

1) Single write transfer

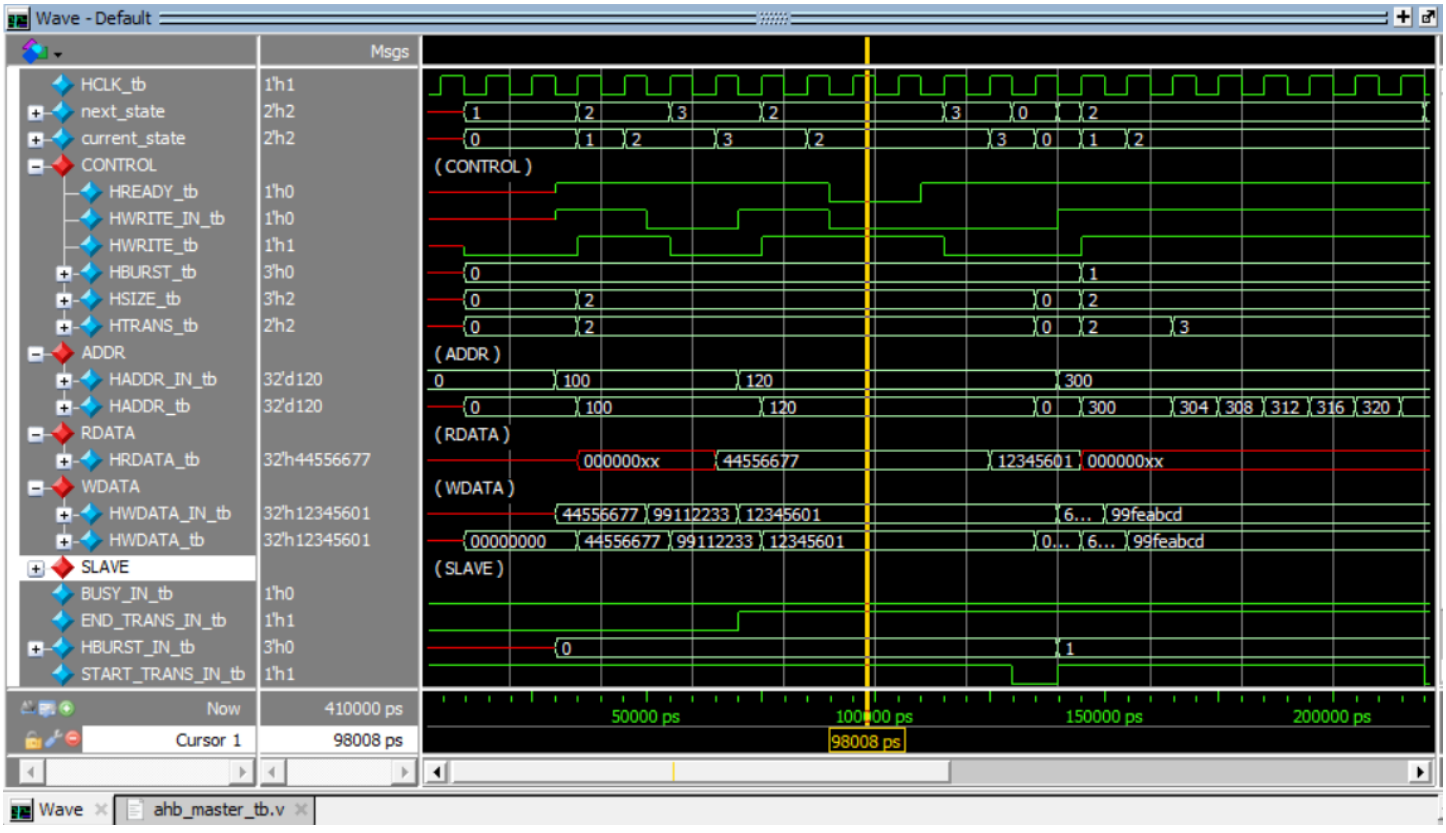




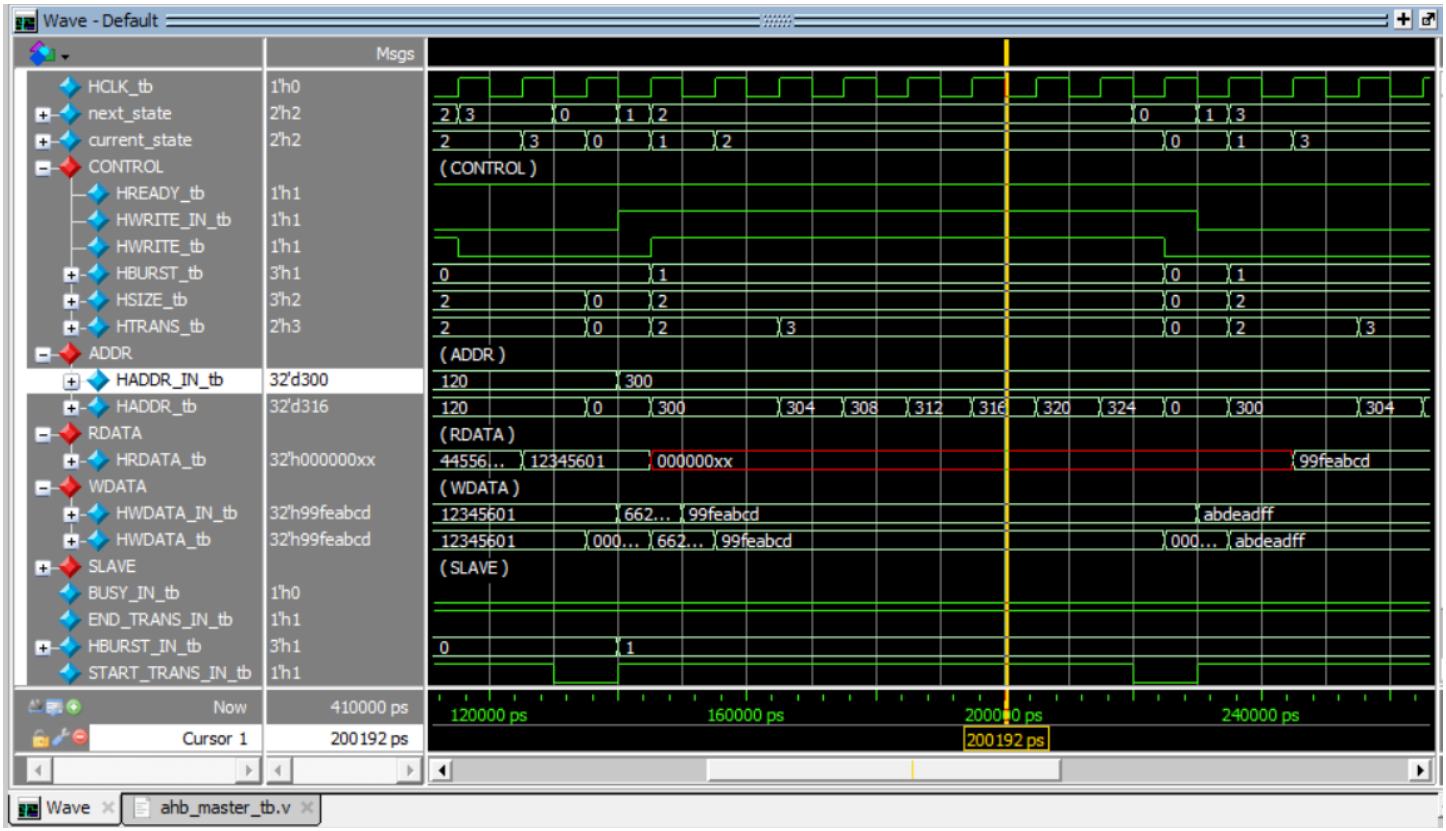
2) Single read transfer

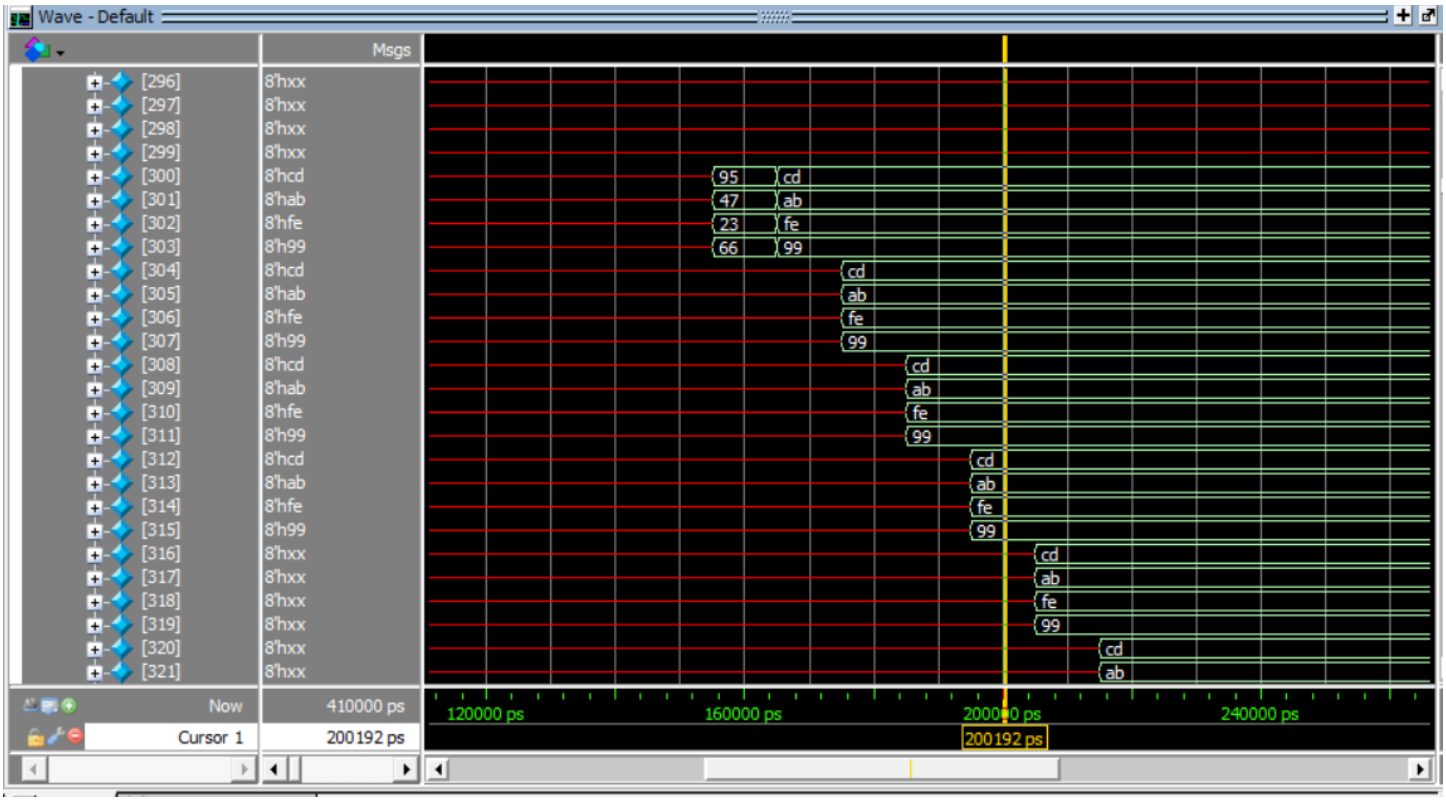


3) Waited transfers

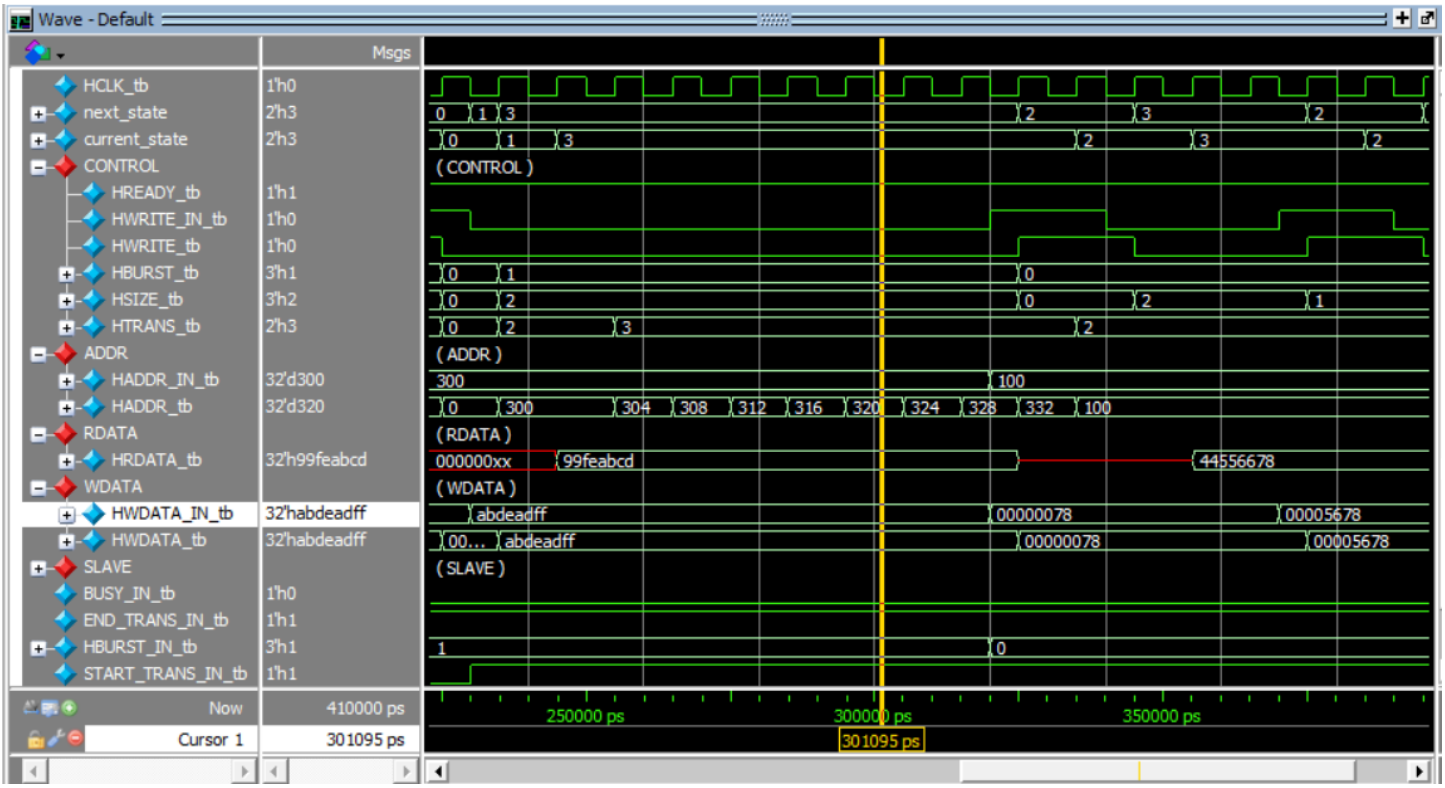


4) INCR burst (write operation)

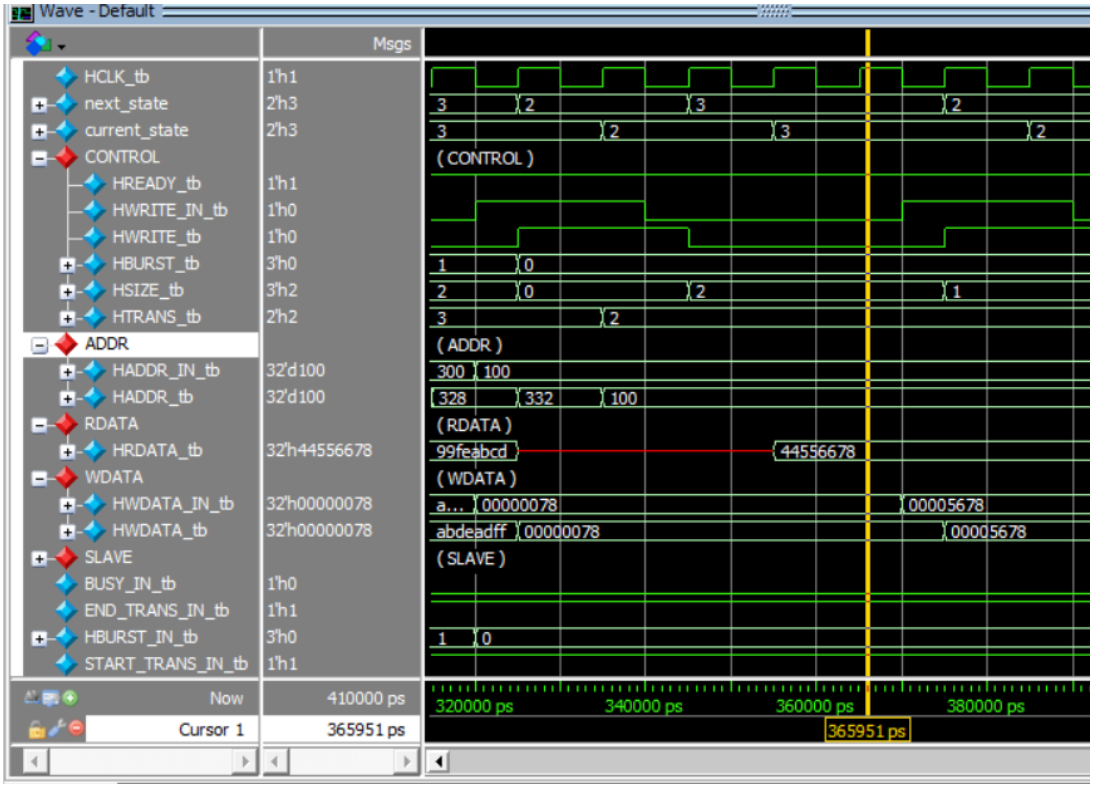


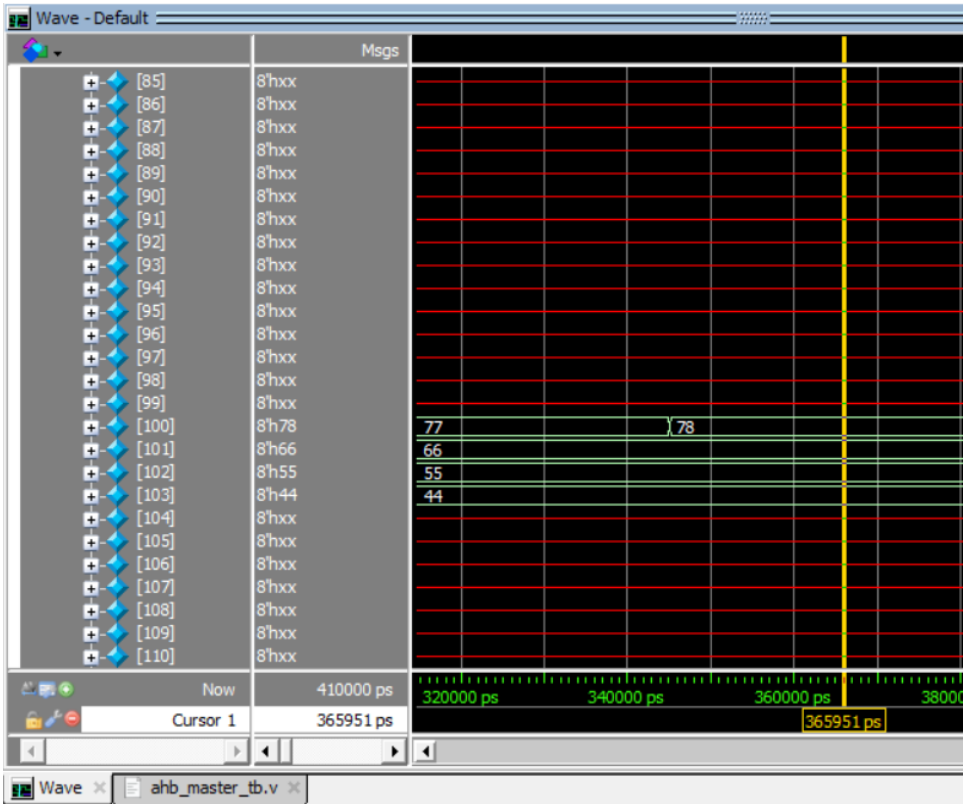


5) INCR burst (read operation)

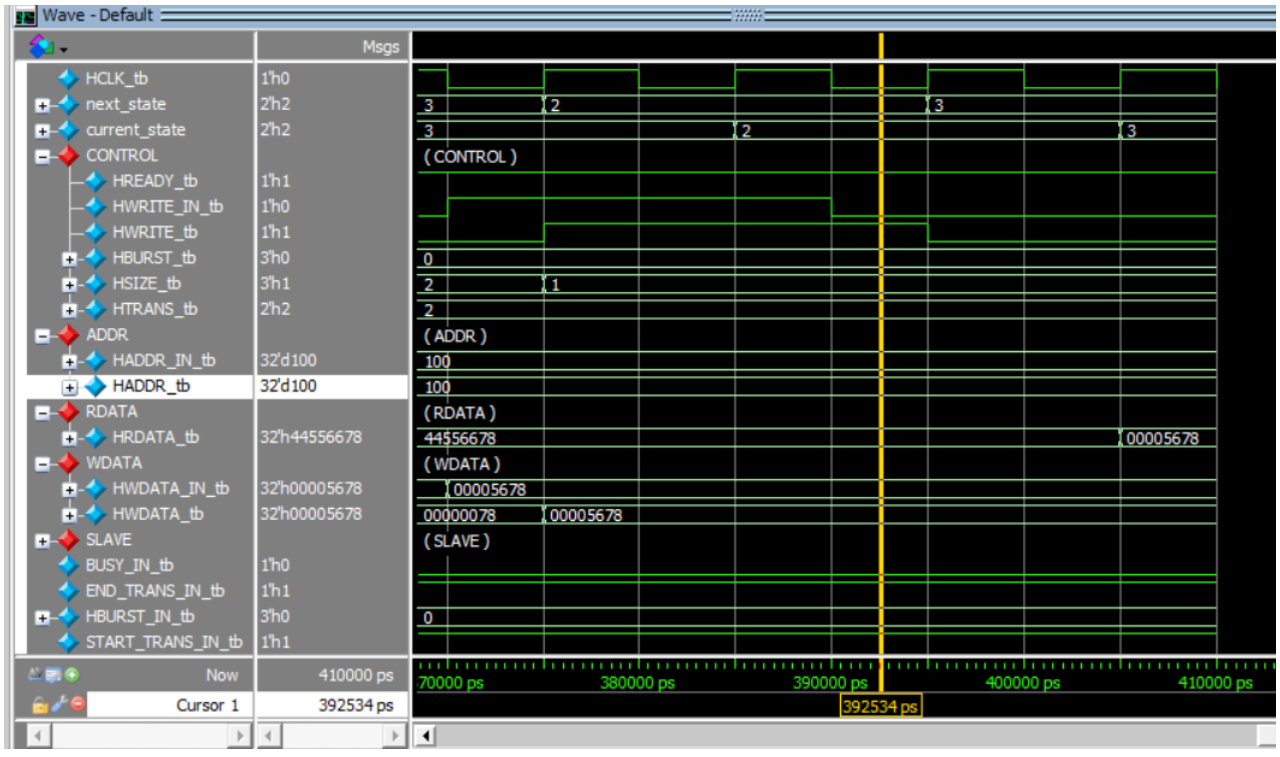


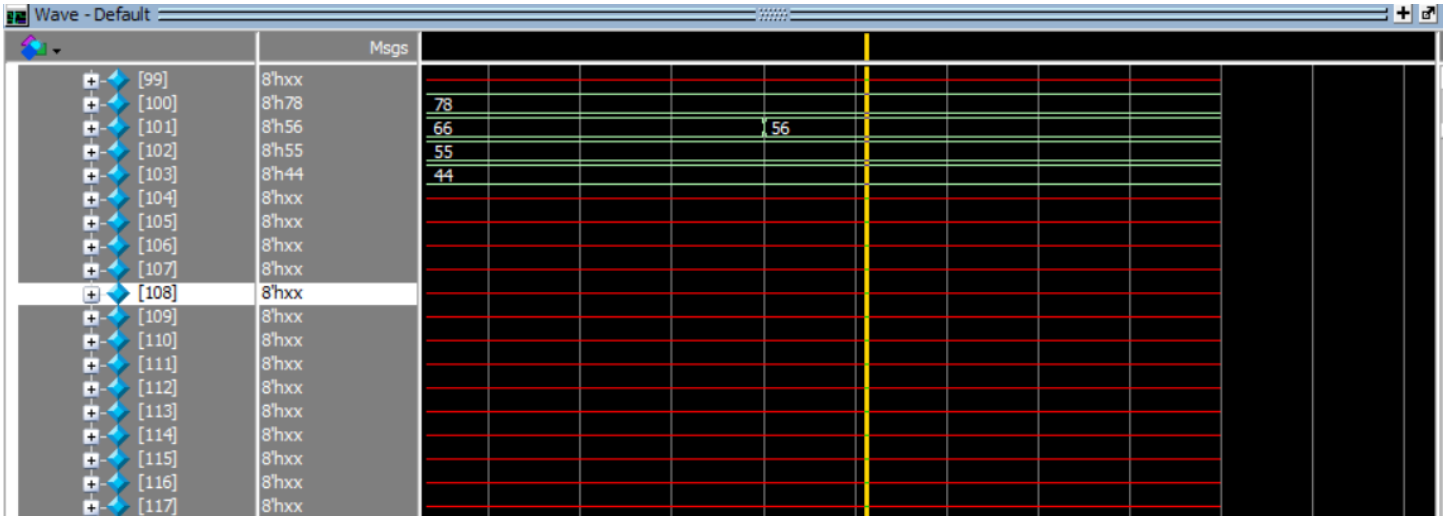
6) Byte write and read





7) Halfword write and read




Wave - Default
Msgs
[99] 8'hxx
[100] 8'h78
[101] 8'h56
[102] 8'h55
[103] 8'h44
[104] 8'hxx
[105] 8'hxx
[106] 8'hxx
[107] 8'hxx
[108] 8'hxx
[109] 8'hxx
[110] 8'hxx
[111] 8'hxx
[112] 8'hxx
[113] 8'hxx
[114] 8'hxx
[115] 8'hxx
[116] 8'hxx
[117] 8'hxx
78
66
56
55
44